3-5-26 (WEEK 2)

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QUESTION

may not be mutually exclusive

Given arbitrary events $E, F, G \subseteq S$, what is $\mathbb{P}(E \cup F \cup G)$?

PROPOSITION

Given any 3 events E, F, G in a probability space $(S, \mathbb{P})$, $\mathbb{P}(E \cup F \cup G) = \mathbb{P}(E) + \mathbb{P}(F) + \mathbb{P}(G) - \mathbb{P}(E \cap F) - \mathbb{P}(E \cap G) - \mathbb{P}(F \cap G) + \mathbb{P}(E \cap F \cap G)$

Proof

Notice, $\mathbb{P}(E \cup F \cup G) = \mathbb{P}((E \cup F) \cup G)$

$$
\begin{aligned}
&= \mathbb{P}(E \cup F) + \mathbb{P}(G) - \mathbb{P}((E \cup F) \cap G) \\
&= \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F) + \mathbb{P}(G) - \mathbb{P}[(E \cap G) \cup (F \cap G)] \\
&= \mathbb{P}(E) + \mathbb{P}(F) + \mathbb{P}(G) - \mathbb{P}(E \cap F) - [\mathbb{P}(E \cap G) + \mathbb{P}(F \cap G) - \mathbb{P}((E \cap G) \cap (F \cap G))] \\
&= \mathbb{P}(E) + \mathbb{P}(F) + \mathbb{P}(G) - \mathbb{P}(E \cap F) - \mathbb{P}(E \cap G) - \mathbb{P}(F \cap G) + \mathbb{P}(E \cap F \cap G) \quad \square
\end{aligned}
$$

EXAMPLE

Consider if we roll a fair 60-sided die numbered 1 to 60.

We win if the number is a multiple of 2, 3, or 5.

What is $\mathbb{P}(\text{win})$?

Solution

We know $S = [60]$ (options of the roll) and since it's a uniform probability, if $E \subseteq S$, $\mathbb{P}(E) = \frac{|E|}{60}$

Here, $W = \{\text{winning numbers}\} = \{x \in S \mid (2 \text{ or } 3 \text{ or } 5) \mid x\}$

We define $E_k := \{x \in S \mid k \mid x\}$ $\forall k \in \mathbb{N}$

$\therefore$ $W = E_2 \cup E_3 \cup E_5$

Consider the prob of subevents

$\mathbb{P}(E_2) = \frac{|E_2|}{60} = \frac{30}{60} = \frac{1}{2}$

$\mathbb{P}(E_3) = \frac{|E_3|}{60} = \frac{20}{60} = \frac{1}{3}$

Notice, if $k \mid 60$, then $|E_k| = \frac{60}{k}$. More generally, $|E_k| = \lfloor \frac{60}{k} \rfloor$

$\therefore$ $\mathbb{P}(E_k) = \frac{\lfloor \frac{60}{k} \rfloor}{60}$ (i.e. $\frac{1}{k}$ if $k \mid 60$)

$\mathbb{P}(E_2 \cap E_3) = \mathbb{P}(E_6) = \frac{1}{6}$

More generally, if $\gcd(m,n) = 1$, then $E_k \cap E_l = E_{kl}$. In general, $E_k \cap E_l = E_{\text{lcm}(k,l)}$

Thus: $\mathbb{P}(E_3 \cap E_5) = \frac{1}{15}$, $\mathbb{P}(E_2 \cap E_5) = \frac{1}{10}$, $\mathbb{P}(E_2 \cap E_3 \cap E_5) = \mathbb{P}(E_6 \cap E_5) = \frac{1}{30}$

$$
\begin{aligned}
\therefore \mathbb{P}(W) &= \mathbb{P}(E_2 \cap E_3 \cap E_5) \\
&= \mathbb{P}(E_2) + \mathbb{P}(E_3) + \mathbb{P}(E_5) - \mathbb{P}(E_2 \cap E_3) - \mathbb{P}(E_2 \cap E_5) - \mathbb{P}(E_3 \cap E_5) + \mathbb{P}(E_2 \cap E_3 \cap E_5) \\
&= \frac{1}{2} + \frac{1}{3} + \frac{1}{5} - \frac{1}{6} - \frac{1}{10} - \frac{1}{15} + \frac{1}{30} \\
&= \frac{11}{15}
\end{aligned}
$$

THEOREM (THE INCLUSION-EXCLUSION PRINCIPLE)

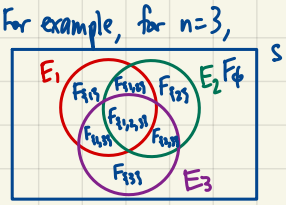
Let $(S, \mathbb{P})$ be a probability space, and let $E_1, \ldots, E_n$ be arbitrary events in S. Then, $\mathbb{P}(\bigcup_{i=1}^{n} E_i) = \sum_{r=1}^{n} (-1)^{r+1} \sum_{1 \le i_1 < \ldots < i_r \le n} \mathbb{P}(E_{i_1} \cap \ldots \cap E_{i_r})$

Alternatively, $\mathbb{P}(\bigcup_{i=1}^{n} E_i) = \sum_{\emptyset \neq I \subseteq [n]} (-1)^{|I|+1} \mathbb{P}(\bigcap_{i \in I} E_i)$

Proof

Define, $\forall I \subseteq [n]$, $F_I := (\bigcap_{i \in I} E_i) \cap (\bigcap_{i \notin I} E_i^c) = \{\text{event } E_i \text{ occurs} \Leftrightarrow i \in I\}$ which by def is mutually exclusive

For example, for $n = 3$,



Observe: $E_1 \cup E_2 \cup \dots \cup E_n = \bigsqcup_{I \subseteq [n], I \neq \emptyset} F_I \Rightarrow \mathbb{P}(\bigcup_{i=1}^n E_i) = \sum_{I \neq \emptyset, I \subseteq [n]} \mathbb{P}(F_I)$

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Furthermore, $\forall I \subseteq [n]$, $\bigcap_{i \in I} E_i = \bigsqcup_{J \subseteq I} F_J \Rightarrow \mathbb{P}(\bigcap_{i \in I} E_i) = \sum_{I \subseteq J \subseteq [n]} \mathbb{P}(F_J)$

$$\therefore \sum_{\emptyset \neq I \subseteq [n]} (-1)^{|I|+1} \mathbb{P}(\bigcap_{i \in I} E_i) = \sum_{\emptyset \neq I \subseteq [n]} (-1)^{|I|+1} \sum_{I \subseteq J \subseteq [n]} \mathbb{P}(F_J)$$
$$= \sum_{\emptyset \neq J \subseteq [n]} \mathbb{P}(F_J) \boxed{\sum_{\emptyset \neq I \subseteq J} (-1)^{|I|+1}}$$

Let $r := |I|$, then $(-1)^{|I|+1} = (-1)^{r+1}$ and the number of such I is $\binom{|J|}{r}$

$$\therefore \sum_{\emptyset \neq I \subseteq J} (-1)^{|I|+1} = \sum_{r=1}^{|J|} \binom{|J|}{r} (-1)^{r+1} = -\underbrace{\sum_{r=1}^{|J|}}_{\text{bin thm } \Rightarrow r=0} \binom{|J|}{r} (-1)^r 1^{|J|-r} = -\left[\sum_{r=0}^{|J|} \binom{|J|}{r} (-1)^r - \binom{|J|}{0} (-1)^0\right] = -\left[(-1+1)^{|J|} - 1\right] = 1$$

$$\therefore \sum_{\emptyset \neq I \subseteq [n]} (-1)^{|I|+1} \mathbb{P}(\bigcap_{i \in I} E_i) = \sum_{\emptyset \neq J \subseteq [n]} \mathbb{P}(F_J) = \mathbb{P}(\bigcup_{i=1}^n F_i) \quad \square$$

EXAMPLE

n people go to a party. Each person takes a random hat. What is the probability that nobody takes their own hat?

Solution

Notice, $\mathbb{P}(\text{everybody gets their own hat}) = \frac{1}{n!}$

Consider if $n=1$, then $\mathbb{P}(\text{someone gets their hat}) = 1$

If $n=2$, $\mathbb{P}(\text{someone gets their hat}) = \frac{1}{2}$

If $n=3$,
1 2 3 ✓ $\Rightarrow$ Prob $= \frac{2}{3}$
1 3 2 ✓
2 1 3 ✓
2 3 1
3 1 2
3 2 1 ✓

Here, our probability space is: $S = S_n$ = permutation of n hats ($= [n]$), $\mathbb{P}$: uniform, $E = \{\text{somebody gets their own hat}\} = \{\exists i \mid \pi(i) = i\}$

Let $E_i := \{\text{person } i \text{ gets their own hat}\} = \{\pi \in S_n \mid \pi(i) = i\}$

Then, $E = \bigcup_{i=1}^n E_i$

By inclusion-exclusion, $\mathbb{P}(E) = \mathbb{P}(\bigcup_{i=1}^n E_i) = \sum_{\emptyset \neq I \subseteq [n]} (-1)^{|I|+1} \mathbb{P}(\bigcap_{i \in I} E_i)$

Consider $|\bigcap_{i \in I} E_i|$, we know that $\pi(i) = i$ $\forall i \in I$, the other values are arbitrarily permuted

As # other values $= n - |I|$, thus $|\bigcap_{i \in I} E_i| = (n - |I|)!$

$\therefore \mathbb{P}(\bigcap_{i \in I} E_i) = \frac{(n-|I|)!}{n!}$

$$\Rightarrow \mathbb{P}(E) = \sum_{\emptyset \neq I \subseteq [n]} (-1)^{|I|+1} \frac{(n-|I|)!}{n!} \overset{\text{all sets of } I \text{ are sym}}{=} \sum_{r=1}^{n} \binom{n}{r} (-1)^{r+1} \frac{(n-r)!}{n!} = \sum_{r=1}^{n} \frac{(-1)^{r+1}}{r!}$$

Observe: $\mathbb{P}(\text{nobody gets their own hat}) = \mathbb{P}(E^c) = 1 - \mathbb{P}(E) = 1 - \sum_{r=1}^{n} \frac{(-1)^{r+1}}{r!} = \sum_{r=0}^{n} \frac{(-1)^r}{r!}$

derangement: nothing ends up where it started

As $n \to \infty$, $\sum_{r=1}^{n} \frac{(-1)^r}{r!} \longrightarrow \sum_{r=1}^{\infty} \frac{(-1)^r}{r!} = \frac{1}{e}$